MILITARY SPECIFICATION

INSTALLATION REQUIREMENTS, AIRCRAFT PROPULSION SYSTEMS, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers the performance, design, and testing of the airframe portion of the propulsion installation system (engine installation) and establishes the technical requirements applicable to the following propulsion installation subsystems and specialty areas:

Engine starting (3.4.1)

Propulsion system controls (3.4.2)

Fire protection (3.4.3)

Drains (3.4.4)

Cooling and ventilation (3.4.5)

Engine exhaust (3.4.6)

Air induction (3.4.7)

Plumbing and wiring installation (3.4.8)

Engine buildups (quick engine change assemblies) (3.4.9)

Accessibility (3.4.10)

Vulnerability (3.4.11)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

TT-P-59 Paint, Ready Mixed, International Orange

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Military

MIL-E-5007	Engines, Aircraft, Turbojet and Turbofan, General Specification for
MIL-E-5009	Engines, Aircraft, Turbojet and Turbofan, Tests for
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-H-5440	Hydraulic Systems, Aircraft, Types I and II, Design, Installation, and Data Requirements for
MIL-F-5442	Fittings, Push-Pull Engine Control, Quick Disconnect
MIL-F-7179	Finishes and Coatings: General Specification for Protection of Aerospace Weapons Systems, Structures and Parts
MIL-C-7413	Couplings, Quick Disconnect, Automatic Shutoff, General Specification for
MIL-F-7872	Fire and Overheat Warning Systems, Continuous, Aircraft: Test and Installation of
MIL-C-7958	Controls; Push-Pull, Flexible and Rigid
MIL-A-8064	Actuators and Actuating Systems, Aircraft, Electromechanical, General Requirements for
MIL-D-8181	Detector, Ice, Air Intake Duct, Aircraft Engines and Airframe Systems, General Specification for
MIL-S-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Accelerator Required
MIL-I-8776	Insulation Blanket, Thermal, Aircraft Gas Turbine Engine
MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-A-9482	Anti-Icing Equipment for Aircraft, Heated Surface Type, General Specification for
MIL-D-17984	Data Presentation Requirements, Installed Engine Performance and Air Induction Systems
MIL-E-22285	Extinguishing System, Fire, Aircraft, High-Rate-Discharge Type, Installation and Test of
MIL-S-23586	Sealing Compound, Electrical, Silicone Rubber, Accelerator Required
MIL-S-25057	Screen System, Air Inlet, Aircraft Turbine Engine, General Specification for
MIL-C-25427	Coupling Assembly, Hydraulic, Self-Sealing, Quick Disconnect
MIL-H-25579	Hose Assembly, Tetrafluoroethylene, High Temperature, Medium Pressure
MIL-T-25920	Test, Ground and Flight, Aircraft Gas Turbine Propulsion System Installation
MIL-C-27536	Coupling, Clamp, Grooved, V-Band
MIL-D-27729	Detecting Systems; Flame and Smoke, Aircraft and Aerospace Vehicles, General Performance, Installation and Test of
MIL-H-38360	Hose Assembly, Tetrafluoroethylene, High Temperature, High Pressure, Hydraulic and Pneumatic
MIL-F-38363	Fuel System, Aircraft, Design, Performance, Installation, Testing, and Data Requirements, General Specification for

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MIL-E-38453	Environmental Control, Environmental Protection, and Engine Bleed Air Systems, Aircraft and Aircraft Launched Missiles, General Specification for
MIL-I-81550	Insulating Compound, Electrical, Embedding, Reversion Resistant Silicone

STANDARDS

Military

MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-203	Aircrew Station Controls and Displays for Fixed Wing Aircraft
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-250	Aircrew Station Controls and Displays for Rotary Wing Aircraft
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for
MIL-STD-889	Dissimilar Metals
MS33602	Bolts, Self-Retaining, Aircraft Reliability and Maintainability Design and Usage Requirements for
MS33611	Tube Bend Radii
MS33790	Bend Data, Minimum, Hydraulic, Pneumatic, Fuel and Oil Hose

PUBLICATIONS

Air Force Systems Command Design Handbook

AFSC DH2-3	Propulsion and Power

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Society of Automotive Engineers, Inc.

Aerospace Recommended Practice

ARP 1055	Fire Resistance and Fire Test Requirements for Fluid System Components

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(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, New York 10001.)

3. REQUIREMENTS

3.1 Subsystem components. The propulsion installation system components shall comply with the applicable military specifications. In those instances where no applicable specification exists, the contractor shall prepare a detail specification. The specification shall be subject to procuring activity approval prior to initiation of component tests.

3.2 Selection of specifications and standards. Selection of specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Design standards. MS, AN, and AND standards shall be used whenever applicable.

3.3 Materials. Materials used in the manufacture of the components and hardware included in the propulsion installation system shall be suitable for the environment expected for the aircraft and shall conform to applicable government specifications or equivalent contractor specifications. The use of contractor specifications must be approved by the procuring activity.

3.3.1 Metals. Metals used in the propulsion installation system shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, oil, salt spray, atmospheric conditions, or wear likely to be encountered in transportation, storage, or during normal service use.

3.3.2 Dissimilar metals. Dissimilar metals used in intimate contact with each other shall be protected in accordance with MIL-STD-889.

3.3.3 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with fungicidal agents acceptable to the procuring activity.

3.3.4 Potting compounds. Encapsulating and embedment applications of potting compound shall comply with requirement 47 of MIL-STD-454. In addition to the requirements of this standard, the following shall apply:

a. The use of electrical connectors requiring potting compound material should be avoided.

b. Materials used for encapsulation and embedment shall be carefully selected for their operating environment condition. Only those materials which meet the requirements of MIL-S-8516, MIL-S-23586, and MIL-I-81550 shall be used. Any departure from or deviation to these specifications must have prior approval.

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3.4 Design and construction. The propulsion installation system shall be designed in accordance with the requirements of this specification and those dictated by the operational environment and mission of the weapon system as described in the weapon system specification. The propulsion installation system shall function satisfactorily under all design operating conditions for the aircraft.

3.4.1 Engine starting. The engine starting system shall consist of all airframe installed equipment, ducting, controls, wiring, etc, which function to provide ground and air starting of the aircraft main engine(s). The power generating, energy storage, energy conversion, and energy transmission means of the starting system shall be considered part of the system when they are installed onboard the aircraft. The system shall be of fail-safe design and shall not compromise personnel safety or the structural integrity of the airframe/engine in any operating mode or environment. Equipment which employ high-speed rotating components (turbines, etc) shall contain within their envelope all fragments resulting from a maximum energy hub burst failure.

3.4.1.1 Starter. The starting system shall provide satisfactory main engine ground starting and inflight starting, when required, under all conditions and environments specified in the applicable aircraft specification. Unless otherwise specified in the aircraft specification, satisfactory ground starts of an installed engine shall be achieved at all points within the envelope shown on figure 1. The system shall accelerate an installed main engine, including all required engine-driven accessories, from rest to the required engine assist speed within the maximum allowable starting times shown on the figure entitled "Engine Ground Starting Time Versus Ambient Air Temperature" in MIL-E-5007. The starting times at other pressure altitudes may vary; however, the total starting time shall be subject to procuring activity approval. In multiengine aircraft the maximum engine start time shall apply to all engine positions. The information in Design Handbook AFSC DH2-3 pertaining to turbine engine starters may be used as guidance in establishing the design and operating characteristics of the starting system. The selected system shall conform to those military specifications applicable to the particular system configuration.

3.4.1.2 Controls and ignition. The starting system controls shall provide the pilot with a means of initiating and terminating the engine start cycle during ground and inflight starting, and for cranking the engine(s) without ignition. After initiation, the system shall be automatically controlled through termination, and appropriate crew station indication shall be provided to ensure that the various controls are inactivated when not required for operation. Controls shall be provided for initiating and maintaining continuous engine ignition system operation during takeoff, periods of inflight turbulence, and during approach/landing. The continuous ignition system shall have the capability to reinitiate combustion following an engine flameout without requiring manipulation of the engine power lever, provided the cause of engine flameout has been removed. If an engine is equipped with an automatic relight system, cockpit indication of

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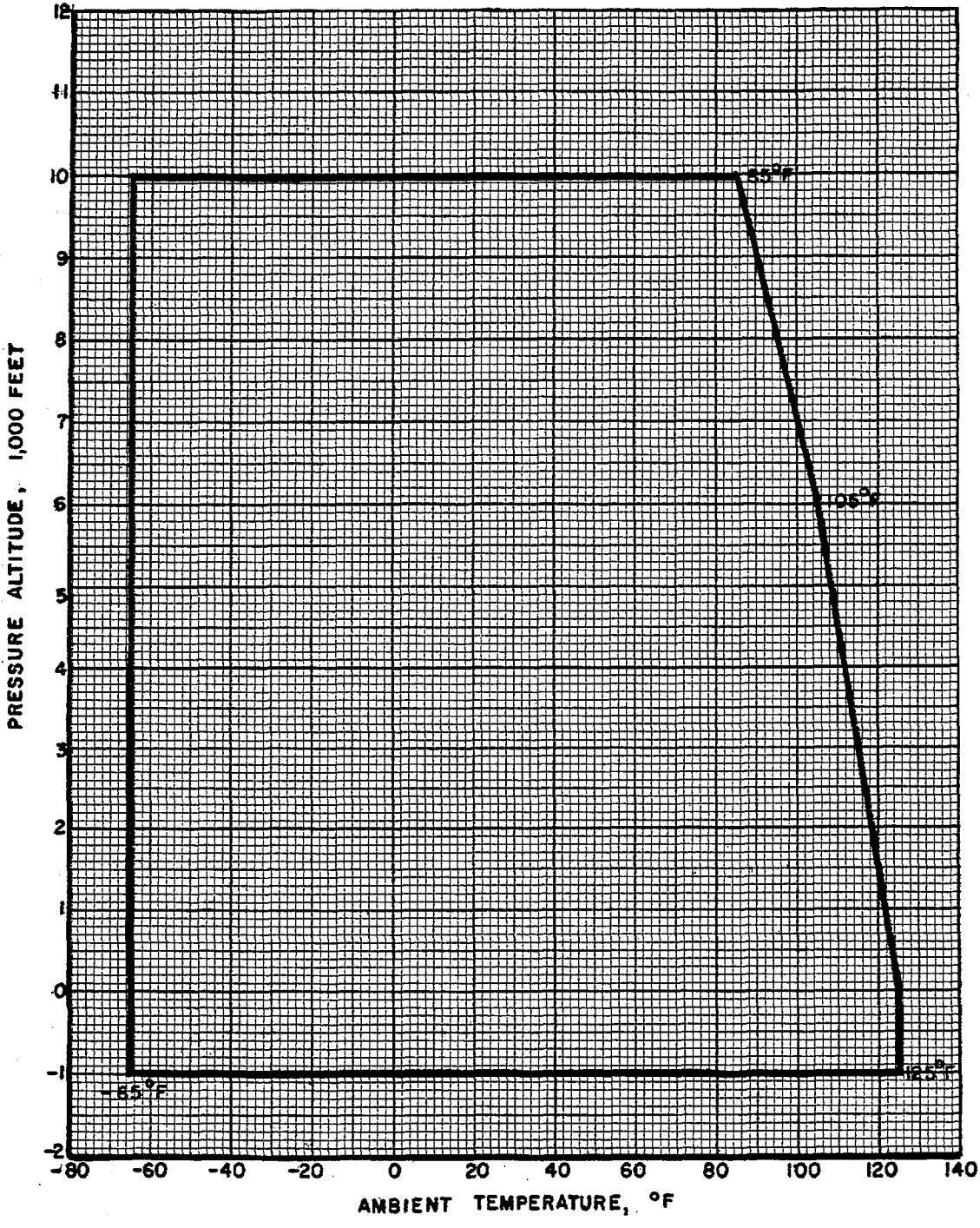


FIGURE 1. Ground Starting Envelope

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system actuation shall be provided. An automatic relight system shall not eliminate the requirement for continuous ignition.

3.4.2 Propulsion system controls. The propulsion system control components covered by this specification shall consist of those airframe-furnished mechanical, pneumatic, hydraulic, and electric linkage mechanisms and related components which connect the cockpit control levers to the engine contractor-furnished controls. The control mechanisms include, for example, those associated with turbojet, turbofan, and turboprop power levers (throttles) and propeller condition levers. The engine contractor-furnished controls are not covered by this specification. The propulsion system cockpit controls shall comply with MIL-STD-203 and MIL-STD-250.

3.4.2.1 Control mechanisms. The control mechanisms may consist of push-pull rods and bellcranks, cables and pulleys, flexible controls, remote-positioning controls, force amplifiers, hydraulic controls, or combinations thereof, provided the functional, maintainability, reliability, serviceability, and fail-safe features are acceptable to the procuring activity. The power lever linkages between the flight station and the engines shall provide a linear kinematic relationship between the positions of the power levers in the flight station and at the engines. The controls shall be positively operated; springs shall not be relied upon to actuate the linkage mechanism in either direction. Antibacklash devices may be used in the control systems; however, upon failure of the control system, such devices shall not cause the linkage mechanism to assume a position which will make the aircraft unsafe or uncontrollable. The control linkages shall be provided with self-retaining bolts conforming to MS33602. Engine control linkages used in fire zones shall be constructed of stainless steel or equivalent fire-resistant material. Flexible and rigid push-pull controls, when used, shall conform to MIL-C-7958 and shall have backlash adjustments at one end of the control cable. Push-pull controls shall have quick disconnects in accordance with MIL-F-5442. In installations where the control systems are extremely long and complex, an electrical control system may be employed; however, specific procuring activity approval shall be obtained. Electromechanical actuators, if used, shall be in accordance with MIL-A-8064. Force amplifiers, when used, shall be fail safe in that an amplifier failure shall result only in the loss of the force amplification feature; the integrity of the control shall be maintained, and the resulting control actuating force shall not exceed the maximum allowable by this specification. The entire control system shall be designed to endure at least 25,000 cycles of operation without failure.

3.4.2.1.1 Control linkage adjustment. A minimum number of adjusting points shall be provided in the control linkages. The adjustment points shall be at the point of mating of any two control elements and the adjustment may be in either the mating or mated element or both. A locking device shall be included in each adjustment point. Linkage components which are not designed

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for service adjustment shall be correctly adjusted at the time of initial assembly and shall be secured to prevent in-service alteration.

3.4.2.2 Control forces. The maximum force required to move the cockpit power lever in a single-engine aircraft shall not exceed 7.5 pounds. In multiengine aircraft, the maximum force required to move all power levers simultaneously shall not exceed 7.5 pounds per engine or 30 pounds for all engines, whichever value is lower. In turboprop installations, the maximum force required to actuate the turboprop condition lever shall not exceed 7.5 pounds per engine. The maximum force on all other controls shall not exceed 10 pounds per control. The forces specified herein shall include both the force necessary to overcome friction in the entire system and the force necessary to actuate the engine element. Movement of the power lever throughout the operating range shall be free of abrupt changes in actuating force. All forces shall be measured at the center of the control lever knob with the friction locking device, if any, in the extreme loose position. The maximum force specified is applicable in either direction of movement and under all conditions of operation, including static conditions with the engine not operating, and at any temperature from +200°F (+95°C) to -65°F (-54°C); however, the maximum breakaway force allowable after a low temperature soak shall not exceed 20 pounds. Force amplifiers, whether electric, hydraulic, or pneumatic, may be used to reduce the operating force provided that upon amplifier failure the control system will continue to meet the force requirements specified herein. Throttle and propeller control systems which do not include means for amplifying the applied force shall be designed such that the movement of each lever while in the operating range shall not exceed 2° in each direction when the control linkage is held rigid at any intermediate position and when a force of 150 percent of the force required to move the control is applied. In control systems which include force amplifiers, the deflection of the control system shall be the amount that the control knob will lead the theoretical position (no deflection) while developing 60 pound-inches of output torque. This deflection shall not exceed 2° in each direction. Cockpit controls which are hand-actuated by push-pull motions shall not move in excess of 0.062 inch from center position in either direction when the control is held rigid at the controlled device and a force of 150 percent of the force required to move the control is applied. The force on the controls shall be applied tangentially at the center of the knob for quadrant controls and in the direction of motion for push-pull controls. The requirements specified above are the maximum allowable on multiengine aircraft and should be materially reduced for single-engine aircraft where the control can be made more direct. Application of the maximum allowable control lever forces shall not subject the controlled component to loads in excess of those specified for the component.

3.4.2.3 Control functional characteristics. The control levers in multiengine aircraft shall occupy the same relative quadrant position for the same control setting throughout the entire engine power range. If automatic

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boost controls are used on multiengine aircraft, the controls shall have a minimum number of adjustment points to enable matching of the power levers to within 0.125 inch throughout the quadrant travel as measured along the profile of the quadrant during advancing and retarding the control. The control quadrant shall contain stops which limit the motion of the control lever. The control linkage shall be rigged such that upon movement of the control lever, the stops which limit the travel of the engine component are contacted prior to contacting the stops on the control quadrant. The control lever shall be designed to contact the quadrant stop upon application of a control lever force of not more than 250 or less than 200 percent of the sum of the forces needed to move the engine component and to overcome the friction in the control system. Control lever cam followers, if used, shall operate within a completely enclosed cam track and shall be replaceable. Control lever position detents shall be used as required.

3.4.2.3.1 Turbojet and turbofan engine controls. A single cockpit power lever for control of each engine shall be installed in turbojet- and turbofan-powered aircraft. This lever shall coordinate all engine variables through a single engine connection. The quadrant shall be configured in accordance with figure 2, item A. The quadrant shall provide an idle stop which will prevent the control from being retracted to the cutoff position without requiring a deliberate offsetting motion to defeat the stop. The control shall be capable of unrestricted advancement from the cutoff position to any forward position. Power lever angular travel shall be 110° ±10°.

3.4.2.3.1.1 Thrust reverser controls. A power lever quadrant in accordance with figure 2, item B, shall be provided for engine installations which incorporate thrust reversers. A lift gate shall be used to permit entrance to the reverse thrust position. A mechanical interconnect shall be included to prevent increasing engine power above idle thrust until the thrust reverser is completely extended. Further travel in the aft direction shall increase reverse thrust by increasing engine power. The full reverse throttle stop shall be adjustable. Control advancement from the reverse thrust position shall be restricted to idle thrust until the thrust reverser is completely retracted. Power lever angular travel shall be 110° ±10°.

3.4.2.3.2 Turboprop engine controls. Two flight station controls for each engine-propeller combination shall be installed in turboprop aircraft; one engine power lever and one turboprop condition lever shall be provided. These controls shall have functional characteristics substantially in accordance with figure 2, items C and D.

3.4.2.3.2.1 Reverse-pitch propeller controls. When reverse-pitch propellers are installed, reverse pitch shall be obtained by movement of the engine power lever past the idle position in the aft direction through the lift point. Motion of the power lever shall energize the reversing circuits of the propeller control. Further travel in the aft direction shall increase reverse thrust by increasing engine power. The quadrant shall include a

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A. TURBOJET AND TURBO-FAN POWER LEVER WITHOUT REVERSE THRUST



B. TURBOJET AND TURBO-FAN POWER LEVER WITH REVERSE THRUST



C. TURBOPROP POWER LEVER



D. TURBOPROP CONDITION LEVER

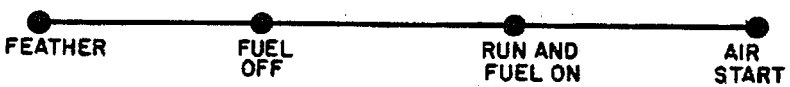


FIGURE 2. Turbine Engine Control Quadrants

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positive stop to prevent inadvertent reverse-pitch operation regardless of the landing gear position. The stop shall consist of an enclosed track having a vertical lift at the flight idle position and shall be bypassed by vertically raising the power lever. Movement of the lever into the aft or reverse position shall not require removal of the operator's hand from the lever. Forward motion of the control shall be unrestricted and shall unreverse the propeller. In multiengine aircraft equipped with two sets of controls, the power lever shall permit lift-to-reverse operation from either pilot's control. The controls shall permit reversal of any propeller independently of the others.

3.4.2.3.2.2 Control lever characteristics

3.4.2.3.2.2.1 Power lever. The power lever shall be in accordance with the standards shown on figure 3. In multiengine aircraft the power lever shall have a minimum vertical lift of 1.125 inches. In single console installations (one set of power levers for both pilots) the maximum upward pull force shall not exceed 1 pound per lever. In dual console installations (one set of power levers for the pilot and a connected set for the copilot) the maximum upward pull force shall not exceed 3 pounds per lever. The vertical lift line shall be at 90° ±10° (maximum) to the plane of the pilot's seat. The upper 2 inches below the knob of each power lever shall be spaced as shown in the lower view on figure 3. The levers within this area shall have rounded edges and a thickness of 0.125 inch.

3.4.2.3.2.2.2 Turboprop condition lever. The condition lever shall be in accordance with the standards shown on figure 4.

3.4.2.3.3 Friction lock. An adjustable friction device shall be incorporated in the throttle control mechanism. The friction device shall permit adjustment by the pilot without requiring removal of his hand from the power lever. The direction of actuation of the device shall be forward, upward, or clockwise to increase friction.

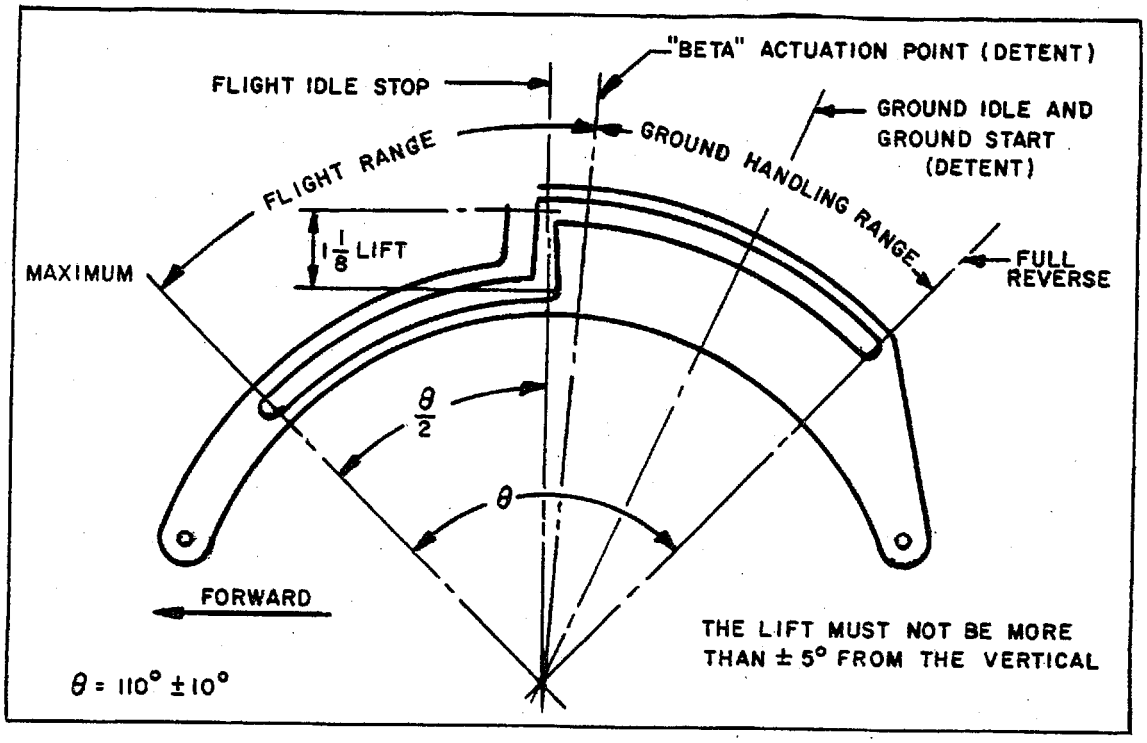
3.4.3 Fire protection

3.4.3.1 Fire and overheat detection. A fire and overheat detection system shall be provided in all zones within the aircraft which are not normally visible to crewmembers and in which fires or overheat conditions can occur.

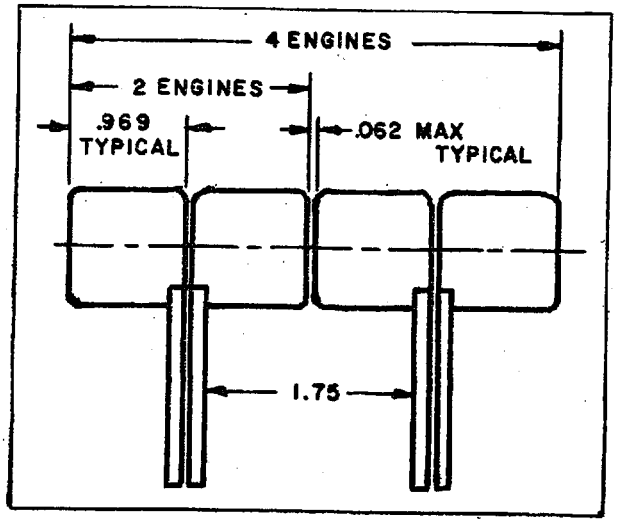
3.4.3.1.1 Fire detection system. Fire detection systems shall be installed in all zones containing combustible fluids and ignition sources and in which a fire or explosion could occur as a result of failure of any system or equipment within that zone. Fire detection systems in accordance with MIL-D-27729 shall be used where fire, and not overheat, is the only hazard.

3.4.3.1.2 Overheat detection system. An overheat detection system shall be installed in all zones where structural components or equipment can be overheated by abnormal engine operation or other failures and where an overheat

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TURBOPROP POWER LEVER MECHANISM



TURBOPROP POWER LEVER ARRANGEMENT
(TWO OR MORE LEVERS)

LINEAR DIMENSIONS IN INCHES (BOTH VIEWS)

FIGURE 3. Turboprop Power Lever

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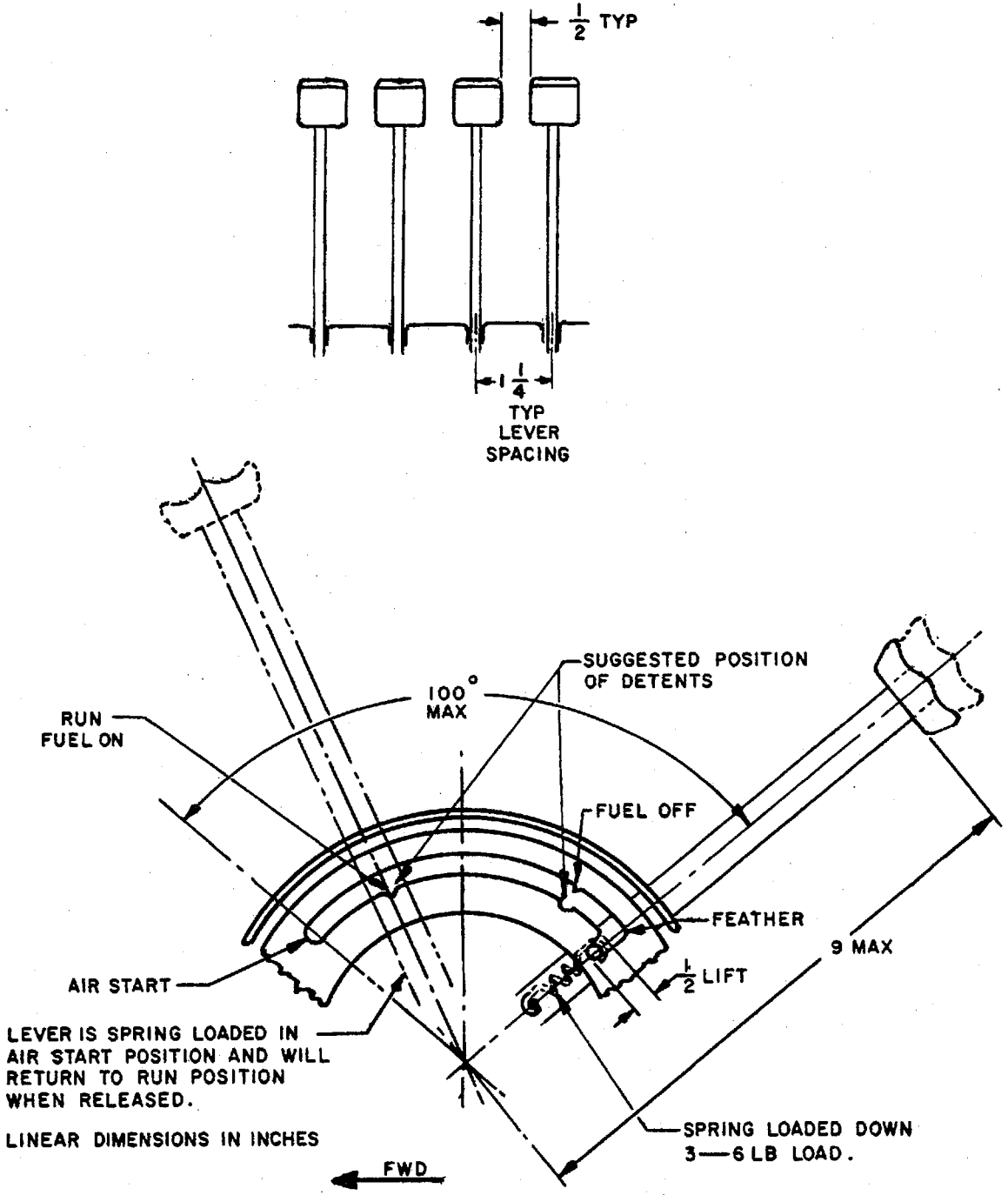


FIGURE 4. Turboprop Condition Lever

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condition will present a flight hazard or add to an existing hazard. Overheat detection systems shall be in accordance with MIL-F-7872.

3.4.3.1.3 Fire and overheat detection system. A continuous-type detection system in accordance with MIL-F-7872 shall be installed in zones which may encounter both fire and overheat conditions. The fire and overheat detection system may be of the integrated type (see 6.3) provided the system is acceptable to the procuring activity.

3.4.3.2 Fire extinguishing system. Fire extinguishing systems shall be installed on all multiengine aircraft. The system shall provide protection for all potential fire zones. Examples of aircraft fire zones are given in the paragraph entitled "Potential Fire Zones" of MIL-E-22285. A system may not be required for a pod engine nacelle where the nacelle pylon isolates the nacelle pod from the remainder of the aircraft; however, specific approval for omitting the extinguishing system shall be obtained from the procuring activity. An extinguishing system shall be installed on experimental single-engine aircraft where the complexity of the engine installation, the inclusion of new and unproven features, or the general importance of the project warrants additional protection for the aircraft. Rotary wing and hover craft with a single engine shall require a one-shot fire extinguishing system.

3.4.3.2.1 Fire extinguishing system installation. Fire extinguishing systems shall be installed and tested in accordance with MIL-E-22285. For extinguishing systems on aircraft where ambient temperatures exceed 200°F or where space is a factor, the nonpressurized, pyrotechnic gas-generator-operated fire extinguisher container shall be considered (see 6.3).

3.4.3.2.1.1 Fire access door. A ground fire access door having minimum dimensions of 5.5 by 10 inches shall be provided for each fire zone. This door shall be in such a position that the nozzle of a ground-type extinguisher can be thrust against the door, forcing it open and permitting the extinguishing agent to be injected directly into the compartment. The retaining device shall be sufficiently strong to hold the door shut against airloads. Quick-release latches shall not be used. The door shall be accessible to ground personnel and shall be located where burning flammable fluid will not drain on the operator of the extinguisher nozzle. The door shall be marked ACCESS FOR FIRE EXTINGUISHER.

3.4.3.3 Isolation of fuel tanks, engine nacelles, and fuselage

3.4.3.3.1 Liquid and vapor barriers. All fuel tanks shall be separated from the engine nacelles, the interior of the wing, and the interior of the fuselage with a liquidproof and vaporproof barrier. The engine installation of all aircraft, regardless of the number or relative position of the engines, shall incorporate a liquid and vapor barrier which separates the accessory and compressor sections or that section of the compartment which contains the greater

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part of the combustible fluid lines and accessories from any engine surface which exceeds a temperature of 700°F. The barrier shall prevent combustible fluid and vapor contact with the hot sections of the engine. If it is not practicable to isolate engine surfaces which exceed temperatures of 700°F, the accessories and drives shall be remotely located and isolated from the engine compartment.

3.4.3.3.2 Firewalls. All fire zones and engines shall be separated from one another, from fuel tanks, and the interior of the wings and fuselage by a fireproof firewall. All firewalls shall be constructed of 0.015 inch thick (minimum) stainless steel or other material which will withstand a 2,000°F flame for 15 minutes. The firewalls shall be liquidproof and vaporproof. Materials used on each face of firewalls, or immediately adjacent thereto, shall be of a type that will not ignite as a result of heat transfer from a flame on the opposite side of the firewall. Adequate spacing shall be maintained between the engine and the firewall in order that a flame from a ruptured burner will not burn through the firewall.

3.4.3.4 Equipment and hardware in potential fire zones. All flammable fluid lines and hose assemblies located in any potential fire zone and engine compartment bleed air lines shall be constructed of stainless steel or other material having equivalent fire resistance; the lines and hose assemblies shall withstand a 2,000°F flame for at least 5 minutes without leakage. The fire-resistant capability shall be maintained when the lines and hose assemblies are exposed to the minimum fluid flow rates, the maximum fluid temperatures, and the normal vibrational environments. Fittings and supports shall have an equal resistance to fire. These requirements shall not apply to vent and drain lines which upon failure will not contribute to any existing fire hazard. All equipments and components which contain flammables and are located in a fire zone shall meet the same fire-resistant criteria. The fire-resistant criteria associated with tube and hose assemblies and other equipments shall not apply to single-engine, high-performance fighter aircraft. The use of hose assemblies shall be minimized; wherever possible, tubing or other more rigid assemblies shall be used. The number and lengths of plumbing in the engine compartment shall be held to a minimum in order to reduce the volume of flammable fluid available in a fire situation. Combustible fluid lines, including vent lines, shall not penetrate any nacelle or engine compartment except the ones they feed. In no case shall combustible fluid lines be routed through air inlet ducts or plenums. No equipments, components, or lines shall be permitted in a fire zone unless they are required for the operation or functioning of equipment contained in the fire zone. The surface temperature of all bleed air lines or ducts located in any fire zone shall not exceed a temperature of 700°F under any operating condition.

3.4.3.5 Firewall shutoff valves. Firewall shutoff valves shall be used in all pressurized lines carrying combustible fluids to any fire zone. Shutoff

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valves need not be provided for drain lines and lines used with a closed-loop fluid system peculiar to a system or equipment within the fire zone; e.g., the integrated drive generator oil cooling system lines between the generator and the heat exchanger. The shutoff valve shall be located as near the fire zone as practicable, but not in the zone. If possible, the valve shall be located on the outside face of the firewall which isolates the fire zone. Electrically operated valves shall not change position during flight as a result of a short circuit, stray voltage, electrical power failure, or normal mechanical forces.

3.4.4 Drains. Drainage of all closed compartments in the engine installation or nacelle and pylon (such as the engine accessory section; spaces enclosing fuel, oil, and hydraulic lines and equipment; and other similar spaces) shall be provided for all normal flight and ground attitudes. Inadvertent liquid spillage and accidents as well as combat aircraft battle damage shall be considered when sizing and locating drains. If, for special types and periods of engine operation, combustible fluid leakage is unavoidable, a suitable drain shall be installed to ensure that the leakage does not impair safe operation of the aircraft. Adequate drainage of engine afterburner ducts shall be provided to prevent accumulation of fuel in the duct while the engine is in its normal nonoperating position.

3.4.4.1 Location of drains. Compartment and equipment drains shall remove normal and accidental leakage of fuel, oil, or hydraulic fluid to a safe location outside the aircraft, away from exhaust system openings. Fluids emitted from drains shall not come in contact with the exhaust gas wake nor shall they impinge on or enter any part of the aircraft under any operating condition. A sufficient number of drain holes shall be provided in pylons, bulkheads, stiffeners, and skin to permit the normal flow of accumulated fluids to collect at low points. Scuppers and drains shall be provided to effectively remove collected fluid from the aircraft. All drains shall be so located that scavenging suction is produced in flight.

3.4.4.1.1 Routing of drains. All drain and vent lines shall be routed to permit fluid to exit from the bottom of the fuselage, nacelle, wing, and pylon and shall pass through resilient grommets in bulkheads and cowlings in order to prevent chafing. Drains may be interconnected provided line sizes are made adequate to ensure proper drainage. Interconnection shall not be permitted where return of any fluid or vapor may create a fire hazard or damage any interconnected components. Specifically, engine combustor drains, pressurizing and dump (P and D) valve drains, and afterburner or tailpipe drains shall not be connected with each other or any other drain.

3.4.4.2 Drain size. Individual drain holes and drain tubing shall have a minimum internal diameter of 0.25 inch. When a single drain opening is used for the entire engine accessory section drain system, it shall be sized to accommodate all leakage including that resulting from equipment failure. The drain opening shall be located in an area where the local airstreams are

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such that drained fluids and vapors will exit freely and will not be driven back into the drained compartment or any other compartment. Drainline materials shall be compatible with the drained fluids and the compartment environment.

3.4.5 Cooling and ventilation. The engine compartment or nacelle cooling and ventilation system shall consist of those systems designed to prevent excessive engine accessory, equipment, supporting structure, and compartment temperatures and the accumulation of explosive or flammable mixtures in or about the engine installation.

3.4.5.1 Air inlet. The cooling and ventilating air intakes shall be so located that fuel, oil, and hydraulic fluid liquids and vapors and engine exhaust gases cannot enter the system. In engine compartments divided by a liquid and vapor barrier, separate cooling air intakes shall be provided for each compartment. The cooling system shall not take air from the engine air inlet duct or plenum. Cooling air intake doors may be provided to achieve complete nacelle or compartment isolation. During a fire emergency, the doors shall close automatically and simultaneously with actuation of the firewall shutoff valves. The air intake(s) location shall not be susceptible to ice accretion. If ice formation is critical to an extent that airflow is adversely affected, the intake shall have suitable ice protection.

3.4.5.2 Compartment airflow. When an engine compartment or nacelle is divided by one or more liquid and vapor barriers, separate cooling and ventilation systems shall be provided for each compartment. There shall be no air exchange between engine compartments. Air from other cooling and ventilation systems may discharge into the engine compartment(s) provided the air temperature is 200°F or less and cannot be contaminated with flammable, corrosive, or explosive agents which may result from normal or accidental leakage throughout any flight attitude or engine operating mode. Airflow used to cool and ventilate any engine compartment shall be discharged overboard and shall not be discharged into any other cooling and ventilation system except that used for exhaust nozzle cooling. Cooling air discharge from accessories such as oil coolers and generators shall be discharged overboard. Air passages, ducts, or intakes shall not pass through the engine firewall, unless the duct or passage is made of stainless steel or equivalent fireproof material and is adequately sealed and protected from damage by fire. Cooling system flow reversals or no-flow conditions shall not be permitted during any ground or flight condition; unidirectional airflow shall be provided under all conditions.

3.4.5.3 Compartment cooling. The cooling and ventilation system shall be of sufficient capacity to ensure that all established temperature limits within the engine installation are not exceeded under any operating condition. The airframe contractor shall establish the maximum ambient temperature environments to which the engine and engine compartment plumbing, wiring, equipment, accessories, and structural components may be exposed during all periods of

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aircraft flight operation, ground operation, and after engine shutdown. In establishing these limits, the airframe contractor shall comply with the temperature limitations of the engine (as established by the engine manufacturer), engine components, and other installed equipment. In establishing post-shutdown temperature limitations, the contractor shall consider that a gas turbine engine together with its usual installation orientation does not lend itself to pronounced convective circulation of cooling air. Post-shutdown cooling may further be restricted due to adverse operating conditions such as blowing dust and salt spray which require that all openings in the aircraft be covered as soon as possible after engine shutdown. Post-shutdown temperature limits shall be satisfied without the need for auxiliary ground cooling equipment.

3.4.5.4 Explosive atmosphere prevention. During all normal aircraft operations the cooling and ventilation system shall prevent the accumulation of explosive or flammable air mixtures within the engine installation.

3.4.5.5 Insulation blankets. Insulation blankets, when used, shall conform to MIL-I-8776. In order to prevent blanket expansion due to pressure differentials, small shielded ventholes shall be provided in the cold side foil in areas least susceptible to fuel exposure. The minimum foil thickness shall be 0.002 inch for blankets used to insulate operating temperatures up to 1,200°F. For higher temperatures and afterburner installations, the minimum foil thickness shall be 0.004 inch. A screen shall be incorporated on the cold side of blankets with foils less than 0.003 inch thick to prevent the loss of insulating material from damaged blankets. A lacing method shall be used which will not decrease the thickness of the blanket due to diametral expansion of the protected component.

3.4.6 Engine exhaust system. The exhaust system shall consist of those components which conduct the combustion gases from the point of discharge at the engine to the point of discharge from the aircraft. The system shall include the fan exhaust and the primary exhaust and all related components as installed by the airframe contractor.

3.4.6.1 Exhaust system construction. The exhaust system shall prevent exhaust gas impingement on the airframe and shall also prevent ingestion of exhaust gases by any operating or nonoperating engine or any other air inlet where equipment or system operation, performance, or service life may be adversely affected. The internal and external duct configuration shall be such that restrictions to duct and cooling airflow will be minimized. The exhaust system shall be compatible with the temperature and pressure environment associated with all engine operating conditions. Variable geometry nozzle systems, when used, shall minimize exhaust gas leakage. Exhaust system components and equipment shall be designed and constructed of materials which are not susceptible to corrosion. The airframe skin adjacent to the exhaust gas plume shall be constructed of corrosion-resistant steel or equivalent. The

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exhaust system shall neither promote nor sustain engine compartment fires. The system shall be reparable by normal weld or other repair procedures available at the aircraft's main operating base.

3.4.6.2 Service life. The service life of the exhaust system shall be based on system application and requirements, but in no case shall the life be less than 5,000 hours. Buckling (wrinkling), warping, separation, cracking, or burn-through of the tailpipe and other exhaust system components shall not occur during the service life.

3.4.6.3 Connections. Exhaust system connections that require disassembly during engine removals shall be in accordance with MIL-C-27536 and shall be of the quick-release type. The coupling shall minimize gas and liquid leakage during all conditions of engine and aircraft operation.

3.4.6.4 Exhaust system cooling and protection. Engine exhaust system cooling or shielding shall be provided as required to comply with 3.4.3.3. In installations where exhaust system cooling is provided by an exhaust ejector, the ejector shall provide uniform distribution of the cooling medium in order to obtain maximum cooling efficiency with minimum flow requirements. The ejector shroud shall be removable from the basic engine tailpipe or afterburner installation to facilitate replacement and inspection of both the shroud and the shrouded component. The shroud shall contain a drain system which shall remove unwanted fluids under all normal ground and flight attitudes; there shall be no internal traps in which combustible fluids may collect.

3.4.6.5 Special equipment. Special exhaust system airframe-mounted equipment such as thrust reversers, infrared suppressors, radar cross-section (reflectance) suppressors, noise suppressors, variable geometry nozzles, afterburners, exhaust deflectors, etc, shall be designed to perform satisfactorily with minimum adverse effect on aircraft and engine performance. The equipment shall be fail safe and shall not upon failure jeopardize safe operation of the aircraft or engine. The equipment when in operation shall not cause the exhaust to impinge on the airframe or to be ingested by any operating or nonoperating engine or any other air inlet where equipment or system operation, performance, or service life may be adversely affected. Equipment operation shall not cause foreign objects to be ingested by the engines. Thrust reversers and exhaust deflectors or attenuators shall be designed such that, in the event of a single failure, the device will remain in or assume the forward thrust (stowed or clean) position with no degradation in aircraft or engine performance.

3.4.6.5.1 Thrust reversers. Thrust reversers shall incorporate a means which ensures positive reverser locking in the stowed position. Flight station indication of an unlocked reverser shall be provided. As a minimum performance objective, the thrust reverser shall have a 50-percent affectivity

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on the engine thrust stream acted upon (fan, gas generator, or both). The engine power rating shall be that installed thrust available from an engine operating at a maximum continuous sea level static condition. The thrust reverser shall be capable of operation from the stowed position to the deployed position in not more than 2 seconds and from the deployed position to the stowed position in not more than 5 seconds during all conditions under which the system must operate. The system shall be designed for a minimum total operating life of 10,000 cycles. A cycle shall consist of reverser operation from the stowed position to the deployed position and return to the stowed position.

3.4.7 Air induction system. The air induction system includes the necessary ducts, scoops, auxiliary inlet doors, passages, chambers, etc, which obtain ambient air and supply this air to the engine. Ice prevention and removal provisions, dust separators, and any other equipment installed in or near the intake ducts for the purpose of controlling, restricting, filtering, heating, or cooling the intake air or supporting structure are also considered part of this system.

3.4.7.1 Location of air inlets. The engine air inlet shall be located to assure that satisfactory airflow patterns, including boundary layer characteristics, are provided to the engine at all attitudes and conditions of operation for which the aircraft is designed. The inlet shall be located to minimize the probability of ingesting airborne foreign objects. In addition, the inlet location shall minimize the ingestion of hot gases and debris generated by the aircraft armament systems which may adversely affect engine or aircraft performance.

3.4.7.2 Miscellaneous design features. The inlet ducting shall be of sufficient strength to withstand all temperatures and pressure differentials encountered under any normal flight or engine condition. The inlet shall also withstand vibrations and structural loads produced by airflow variations and those produced by the engine, thrust reverser, airframe, and armament systems. Riveting in the inlet duct internal surfaces shall be of the flush type. The engine air intake system shall not contain components which may vibrate loose and enter the engine.

3.4.7.3 Performance. The air induction system shall provide total pressure variations at the engine compressor face within the limits established in the applicable engine specification. The system shall also provide engine/airframe compatibility sufficient to ensure satisfactory aircraft performance under all ground and flight conditions. The requirements for presentation of the air induction system performance characteristics shall be substantially in accordance with MIL-D-17984.

3.4.7.4 Inlet control systems. Airflow control by means of variable inlet spikes, plugs, wedges, bypass doors, panels, suck-in doors, etc, may be necessary on high-speed aircraft in order to satisfy engine airflow requirements

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over the complete range of operating conditions. Inlet controls, when used, shall have an accuracy of regulation commensurate with the steady-state and transient characteristics of the basic engine fuel control. The reliability of the inlet control shall be equal to that of the basic engine control and the overall system requirements. The control systems shall include manual and automatic safety features which have been demonstrated to be reliable for providing emergency operation under failure conditions; however, the system need not contain devices which protect against the simultaneous failure of two control system parts, except where the first failure can cause the second. In any case, the inlet control system shall, where practicable, incorporate the following fail-safe features:

a. A failure during takeoff of any single functional part shall not reduce the total thrust below 95 percent of that available.

b. A failure of any single part shall not cause the engine or airframe to exceed any structural operating limit to the extent that failure cannot be prevented by simple corrective action.

c. A failure of any single part shall not cause an abnormal operating condition such that aircraft controllability cannot be maintained by simple corrective action.

3.4.7.5 Auxiliary inlet equipment. Auxiliary inlet equipment such as inlet doors shall operate separately from other inlet equipment and control systems. Inlet doors upon activation or failure shall not act as a source of foreign objects or propagate a level of distortion which could cause engine damage. If necessary, the inlet door system shall incorporate ice protection.

3.4.7.6 Inlet protection systems

3.4.7.6.1 Foreign object damage (FOD) protection. The contractor shall determine the potential of the engine inlet system to ingest foreign objects. If foreign object ingestion is probable under any operating condition, the contractor shall evaluate the engine's ability to ingest foreign objects. Foreign object ingestion which results in any of the following conditions is unacceptable:

a. A catastrophic engine failure

b. An immediate decrease in engine power which renders the aircraft uncontrollable

c. A gradual decrease in engine power which is unregainable and prohibits the aircraft from accomplishing its intended mission

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d. A predicted engine removal or overhaul rate in excess of that normally established as being economically and operationally acceptable.

3.4.7.6.1.1 FOD protection system. When unacceptable foreign object ingestion cannot be prevented by inlet design and location or aircraft operating restrictions, the aircraft shall be equipped with engine or airframe-mounted retractable inlet screens or an equivalent FOD protection system. In all cases, however, VTOL, VSTOL, and STOL aircraft intended for operation from unprepared or primitive grass or dirt landing areas shall be equipped with an effective FOD protection system. When FOD protection is provided by a retractable inlet screen system, the screen shall be composed of aerodynamically shaped vanes and shall be in accordance with MIL-S-25057. A study shall be conducted of each new screen installation in order to define the optimum screen size and configuration in terms of protection provided, engine performance loss, inlet distortion effects, engine FOD tolerance, and screen performance consisting of debris retention capabilities, icing tendencies, self-cleaning capability, strength, and fatigue resistance.

3.4.7.6.2 Sand and dust protection. Engine inlets, which are located such that large quantities of sand and dust may enter the inlet, shall be designed or equipped to prevent engine ingestion of erosive particles which could cause accelerated engine wear and performance loss. Incorporation of inlet design features and equipment of this type shall normally be required for VTOL, VSTOL, and STOL aircraft which are designed to operate from unprepared or primitive grass or dirt landing areas. Specific design requirements for sand and dust protection devices shall be as follows:

a. System efficiency at a dust concentration of 0.0015 gram per cubic foot shall be 80 percent or greater for particles of the 5-50 micron size and 100 percent for particles above the 50 micron size.

b. System total pressure loss shall be not greater than 0.5 percent during operation. Total engine power loss shall not exceed 1 percent during system operation. No power loss shall be allowed when the system is in the bypass mode.

c. The specific weight of the installed system shall be not greater than 0.02 pound per engine horsepower.

d. The system shall be capable of satisfactory operation at all applicable engine airflows.

e. The system shall have a small frontal area in order to minimize drag.

f. The system shall be self-cleaning, easily accessible for repair and replacement of parts, and shall contain adequate ice protection if susceptible to ice accretion.

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3.4.7.6.3 Ice protection. The entrance to the engine air induction system of all aircraft shall be protected against ice formation. Other airframe inlet components such as duct dividers, abrupt duct bends, boundary layer bleed wedges, bypass duct lips, engine duct accessory covers, etc, shall also be protected as necessary. The inlet ice protection system shall be designed to prevent any inlet component from accumulating ice formations which could adversely affect engine operation or which, upon separation, could constitute a potential source of engine FOD or other airframe damage. The system shall be designed to provide satisfactory operation during exposure to the environmental conditions defined by the figures entitled "Continuous Maximum Icing Conditions" and "Intermittent Maximum Icing Conditions" of MIL-E-38453 and the tables entitled "Sea Level Anti-icing Conditions" and "Altitude Anti-icing Conditions" of MIL-E-5007. Cockpit indication of satisfactory system operation including overheat indication shall be provided. Any proven protection philosophy or technique may be used in the design of the system in order to provide the protection required; however, expendable-fluid-type systems may be used only when other methods are not feasible and such systems shall require specific procuring activity approval. Ice protection systems shall be either of two types: anti-icing or deicing.

3.4.7.6.3.1 Anti-icing systems. Anti-icing systems shall during operation prevent the accumulation of ice on the protected surface. The system may operate continuously or intermittently. Systems which operate continuously should, if practicable, use waste heat or heat that is normally lost. Systems which during operation cause a performance loss shall operate only when an icing condition exists. An anti-icing system may run "wet" or run "dry" depending upon the surface being protected. A system which runs "wet" shall maintain a surface condition which will melt all precipitation which impinges on the surface and will prohibit the resulting liquid from freezing on the surface. A system which runs "dry" shall maintain a surface condition which will evaporate all precipitation which impinges on the surface. Engine inlet duct lips shall incorporate a system designed to run "dry" as this system will normally prevent hazardous runback icing within the inlet duct.

3.4.7.6.3.2 Deicing systems. Deicing systems shall permit the accumulation of ice on the protected surface and upon operation shall cause the accumulated ice to be removed from the surface. Hence, deicing systems are cyclic systems and their rate of operation depends on the rate at which a hazardous quantity of ice accumulates on the protected surface. Hot air cyclic deicing systems shall comply with the applicable portions of MIL-A-9482.

3.4.7.6.3.3 Controls. All continuous anti-icing systems and deicing systems shall be provided with an automatic and a manual control system. When operating in the automatic mode, the system shall operate upon receiving a signal from an ice detector.

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3.4.7.6.3.4 Ice detector. An ice detector conforming to MIL-D-8181 shall be provided on all aircraft. The detector shall be installed in the engine air inlet duct in an area conducive to ice formation. In addition to controlling the inlet ice protection system, the detector shall upon generating a signal cause a cockpit icing warning light to illuminate.

3.4.8 Plumbing and wiring installation. Hydraulic and fuel system plumbing and electrical wiring installations shall conform to MIL-H-5440, MIL-F-38363, and MIL-W-5088, respectively.

3.4.8.1 Choice of tubing and hose. Tubing and hose assemblies which carry combustibles throughout the aircraft including the engine compartment shall comply with MIL-H-5440 and MIL-F-38363. In addition, tubing and hose assemblies used in the engine compartment and all other fire zones shall also comply with 3.4.3.4. Aluminum alloy tubing may be used for lines carrying combustibles external to the engine compartment and other fire zones.

3.4.8.2 Installation. All piping shall be routed as directly as practicable, considering vulnerability, accessibility, and compartment obstruction. The positioning of combustible fluid lines relative to engine components and engine bleed air ducts shall minimize the fire hazard associated with fluid leakage through accident and battle damage. Plumbing and wiring shall be routed such that no lines pass unnecessarily close to high temperature components. Fuel, oil, and hydraulic lines and equipment shall be located to minimize the possibility of leaking fluid coming into contact with electrical equipment through either the effect of gravity, airflow, or battle damage. All combustible fluid lines shall be routed below electrical equipment and wires. All fluid lines shall be routed to minimize interference with removal or adjustment of engine accessories and in the most efficient manner to obtain maximum protection from the aircraft structure against battle damage. Where practicable, fuel system plumbing shall be routed through fuel tanks in order to minimize fuel losses and fire hazards resulting from combat damage or fitting failure. Combustible fluid lines shall not be used to support other components, plumbing, wiring, etc. The components and systems shall be provided with integral or built-in fluid lines when practicable. Fluid lines shall be free from sharp bends and restrictions. Access for adequate maintenance and inspection of lines and fittings shall be provided without necessitating engine removal.

3.4.8.2.1 Tubing. Tubing shall be fabricated and installed such that forcing, bending, or stretching is not necessary to accomplish installation. No tube assembly shall be installed or secured in a stressed condition. Metal tubing shall be installed such that vibration or deflection of the aircraft structure to which it is attached will not cause failure of the tubing or any of its connections and supports. Metal tubing shall be supported in accordance with MIL-H-5440 and MIL-F-38363 and shall comply with the minimum bend radii specified in MS33611. Flexible connections shall be used on all fluid lines from the aircraft structure to the engine. Adequate spacing shall be maintained

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between all tubing and between the tubing and the aircraft structure to prevent chafing during any static and flight conditions and as a result of system pressure pulses. All tubing shall be protected against chafing by the use of resilient grommets or protective coverings in those areas having inadequate clearance. Where clamps or blocks are used for tubing support, resilient material shall be used between the clamp or block and the tube. The resilient material shall be compatible with the environment; the effects of temperature and exposure to fuel, oil, and hydraulic fluid shall not cause material deterioration.

3.4.8.2.2 Hose. Hose assemblies shall be in accordance with MIL-H-8795, MIL-H-25579, or MIL-H-38360, as applicable. Installed hose assemblies shall meet the following criteria:

a. Hose assemblies shall consist of hose and end fittings in accordance with requirements of applicable specifications and standards. Hose clamp beaded tube type installations shall not be permitted.

b. Hose installations shall be so designed that normal motion does not impose torsional deflection under any condition of operation which will loosen the hose connecting fittings.

c. Hose assemblies carrying flammable fluids in a potential fire zone shall meet the requirements of 3.4.3.4.

d. Unless otherwise specified in the hose specification, hose installation and operational bend radii shall conform to MS33790.

e. Excessive motion shall be restrained where necessary to prevent chafing or misalignment; however, the hose shall not be rigidly supported except at the end fittings.

f. Hose assemblies shall be selected and installed so that elongation or contraction under pressure, within the hose specification limits, will not be detrimental to the installation, either by causing added strains on the end fitting or by binding or chafing of the hose.

g. Hose assemblies shall have an appropriate length tolerance. The tolerance, both maximum and minimum, shall be considered and designed into the installation.

h. Hose assemblies which have permanent fittings shall be installed with adequate installation and wrenching clearances to permit replacement with a hose assembly which has replaceable fittings.

i. Hose assemblies shall be supported in accordance with MIL-H-5440 and MIL-F-38363.

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3.4.8.2.3 Fittings. Permanent line joints shall be used at all points where line separation is not required to facilitate production or to permit normal aircraft servicing. Connections shall be installed to facilitate servicing and simplify maintenance; however, all lines shall have a minimum number of fittings. Fittings conforming to MS or AN standards shall be used where suitable. Where military standards are not suitable, separable line joints shall meet the following requirements:

a. The joint shall be leakproof throughout the environmental range of its application.

b. The joint shall accommodate practical variations in material and handling techniques encountered in service.

c. The joint shall permit repeated assembly and disassembly without attrition of components.

d. Joint effectiveness shall not deteriorate during the useful life of the airframe.

e. The joint design shall minimize the probability of improper joint assembly or installation.

f. The design shall be of maximum simplicity.

g. The joint shall permit maximum ease of installation and removal without use of special tools. (Note: Tools which are listed in federal stock lists, such as AN standard spanner wrenches, strap wrenches, torque wrenches, and deep sockets, are not to be considered special tools.)

h. The joint shall not require critical torque techniques for proper installation.

i. The joint shall be compatible with the plumbing member to which it is attached.

j. The joint shall not be sensitive to cold flow, swell, or shrinkage of synthetic components.

k. The joint shall not be subject to torque relaxation.

l. The joint shall accommodate established structural requirements plus any self-imposed loads while maintaining those safety factors presently imposed on existing standard fittings.

3.4.8.3 Plumbing and wiring installation specification. When specified, the contractor shall prepare a general plumbing and wiring installation

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specification (see 6.2.c). The specification shall contain the standards, rules, and criteria used by the contractor in the design, mockup, fabrication, testing, installation, and inspection of all aircraft combustible fluid carrying lines and hose assemblies, bleed air lines, and wiring assemblies and their associated fittings and supports. If military standards and specifications containing applicable installation criteria and requirements cannot be satisfied, the plumbing and wiring installation specification shall contain appropriate documentation for each instance of noncompliance. All deviations from military specifications and standards shall be subject to procuring activity approval.

3.4.9 Engine buildups. The engine buildup (EBU) or quick engine change (QEC) assembly shall consist of the engine and those equipments and accessories attached to or which form a part of the engine and its installation which can be easily removed or installed as a complete built-up assembly.

3.4.9.1 Interchangeability. Multiengine aircraft EBUs shall be completely interchangeable between engine positions without physical or electrical modifications to any part of the EBUs and without resorting to EBU sectioning.

3.4.9.2 EBU removal and installation. The time required to remove an EBU in operating condition and to install another EBU and put it in operating condition shall not exceed 30 minutes. In turboprop installations, if the propeller is not part of the EBU, the elapsed time shall include removal of the propeller from the installed engine and installation of the same propeller on the replacement engine. No special tools other than an engine trailer, a propeller sling or fixture, and special propeller tools shall be required for an EBU exchange. The EBU shall be installed such that it can be readily removed as a unit without detaching any basic engine components or accessories directly mounted on the engine or without removing any significant parts of the aircraft structure.

3.4.9.3 Disconnects

3.4.9.3.1 Type. Quick-disconnect mechanical and electrical couplings shall be used at all connections between the EBU and the airframe. Quick-disconnect fluid line couplings shall be in accordance with MIL-C-7413 and MIL-C-25427. All fluid lines passing through firewalls shall have flared-tube bulkhead fittings or equivalent. Fluid couplings of the quick-disconnect type which are used at firewalls shall have a bulkhead fitting end for installation at the firewall.

3.4.9.3.2 Location of disconnects. The disconnects shall be arranged to preclude inadvertent reversal, mismatch, or improper connection of electrical and mechanical connectors, couplings, and fittings. Disconnects shall be located in convenient groupings on a wall of the engine compartment and shall be so arranged that several mechanics may simultaneously work in the area with a minimum of interference.

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3.4.9.3.3 Marking. Disconnects for EBUs shall be marked with a solid band of international orange paint conforming to TT-P-59.

3.4.9.4 Engine mounts. Engine mounts which have common bolted connections shall have a hole clearance of not more than 0.032 inch. When mounting points are of the self-aligning type, ball and socket or tapered joints may be used. The boltholes of self-aligning joints may be 0.094 inch greater than the bolt diameter to allow misalignment of the joint halves during bolt insertion. The two halves of a self-aligning joint shall upon tightening draw the two halves of the joint into alignment such that all shear loads are taken by the joint rather than the bolt. Vibration isolators, when required, shall be incorporated in the engine mounts and shall prevent transmittal of detrimental vibrations from the engine to the airframe or vice versa.

3.4.9.5 Engine cowling. Engine or EBU cowlings shall be provided in order to permit easy access to and rapid removal of the engine or the EBU. When practicable, cowlings shall be hinged to permit rapid access to the engine. Where appropriate, cowling sections may also serve as workstands for maintenance personnel. The cowlings shall have no sharp corners which may be hazardous to personnel. Cowlings designed for removal from the aircraft shall be of reasonable dimensions and weight for ease of handling by ground personnel. In multiengine aircraft the cowlings shall be interchangeable between engines. The cowling design life shall be the same as that of the basic airframe. In the event of a high-pressure bleed air line rupture, the cowlings shall not separate from the nacelle nor shall they experience structural damage which would require a major repair or result in a situation which would impair safe operation of the aircraft.

3.4.10 Accessibility. All equipment in the engine installation shall be located and mounted to permit access for rapid equipment testing, servicing, removal, inspection, and replacement. The design shall permit unobstructed adjustment over the entire range of component movement. Inaccessible and complex structure designs shall be avoided in order to minimize repair time. The effort required to provide accessibility shall be commensurate with the relative frequency and man-hours of inspection, servicing, and repair required on the equipment. The arrangement of equipment and components shall be such that maintenance personnel can accomplish their work without assuming awkward positions or requiring excessive movement from position to position and such that fluids and dirt are unlikely to drop on personnel and on other components during maintenance. Assemblies which are subject to frequent removal shall be attached by bolts rather than by permanent attachments using rivets. Where practicable, quick-disconnect features shall be used. Tool clearance for bolted connections shall be provided. In remote areas and where the use of tools is restricted, mounting brackets with self-locking nut plates may be used provided the nut plates are compatible with the environment. Sufficient access shall be provided to all lubrication fittings to permit the use of standard USAF lubrication equipment without the need for adapters. The

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installation shall be arranged to permit inspection, cleaning, adjustment, removal, and replacement of components and accessories with tools normally contained in a mechanic's tool kit; special tools shall not be bequired.

3.4.10.1 Inspection, cleaning, and adjustments. All accessories and components of the engine installation shall be accessible for inspection, cleaning, and adjustment while installed in the aircraft. Accessibility shall not require removal of the engine, accessory gearbox, propeller, propeller gearbox, transmission, fluid tanks, fluid lines, or important parts of the aircraft structure. Service items accessible on a bare uninstalled engine shall be accessible on the installed engine. Examples of items which shall be accessible are as follows:

a. Variable exhaust nozzle and control assembly

b. Variable inlet geometry components, controls, and sensors

c. Oil pressure relief valves

d. Oil tanks

e. Oil filters

f. Oil coolers

g. Fuel boost pumps

h. Afterburner fuel pump

i. Fuel filters

j. Fuel pressure relief valve

k. Fuel nozzles

l. Fuel control (engine and afterburner)

m. Fuel flowmeter

n. Automatic controls and actuators

o. Actuator motor brushes

p. Sump plugs and drain valves

q. Magnetic drain plugs

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r. Ignitor plugs

s. Ignition excitors

t. Fire detection control units

u. Fire extinguisher spray rings

v. Fire extinguisher bottles

w. Engine mounts

x. Gearbox mounts

y. Propulsion control linkages

z. Throttle boxes

aa. Ice detector and controls

bb. Inlet screen components and controls

cc. Inlet duct anti-icing components and controls

dd. Thrust reverser components and controls

ee. Bleed air ducting and valves

ff. Engine power takeoff (PTO) shaft and couplings

gg. Engine-mounted propeller governor

hh. Propeller unfeathering accumulator

ii. Engine borescope inspection provisions

jj. Engine thermocouples and harness.

3.4.10.2 Removal and replacement. As a minimum, the following accessories and components shall be readily removable and replaceable without removing an engine, accessory gearbox, propeller, propeller gearbox, transmission, fluid tank, fluid line, or important parts of the aircraft structure:

a. Ignitor plugs

b. Ignition excitors

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c. Starter

d. Generators/alternators

e. Constant-speed drives

f. Tachometer generator

g. Oil pressure relief valves

h. Oil tanks

i. Oil filters

j. Oil coolers

k. Oil pumps

l. Oil temperature control valve

m. Fuel boost pumps

n. Afterburner fuel pump

o. Fuel filters

p. Fuel nozzles

q. Fuel control (engine and afterburner)

r. Fuel flow divider

s. Fuel flowmeter

t. Fuel heater

u. Water methanol tank or thrust augmentation fluid tank

v. Automatic controls and actuators

w. Hydraulic pump

x. Tailpipe extension

y. Drain valves

z. Fire detector controls

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aa. Fire extinguisher bottles

bb. Throttle boxes

cc. Ice detector and controls

dd. Inlet screens, components, and controls

ee. Variable inlet geometry components, controls, and sensors

ff. Thrust reverser components and controls

gg. Exhaust nozzles and plugs

hh. Inlet duct lip anti-icing system valves, temperature sensors, and controls

ii. Bleed air ducting and valves

jj. Engine-mounted propeller governor

kk. Propeller unfeathering accumulator

ll. Engine PTO shaft and couplings

mm. Power turbine governor

nn. Engine thermocouples and harness.

3.4.11 Vulnerability. The vulnerability of critical propulsion components and accessories to threats commensurate with the aircraft mission shall be considered in the configuration of the propulsion system installation. To the maximum extent practicable, the accessories and components within the engine installation shall be arranged or protected to ensure that the aircraft has the capability to return to its home base following a single hit from any weapon up to and including 23 millimeters. Fluid lines and essential components and wiring shall be located to benefit from the inherent protection provided by nacelle structure and other less essential components and equipments. Redundant fluid lines, wiring, and components, when used, shall be separated by a minimum of 18 inches. The separation distance need not be maintained if the redundant lines, etc, are separated by structure or other nonessential components which prevent the loss of both lines, etc, due to a single hit. When required, further guidance concerning engine installation vulnerability may be obtained from the procuring activity.

3.4.12 Natural and induced environmental extremes. Equipment and systems in the engine installation shall function satisfactorily during and after exposure

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to the applicable worldwide environmental extremes defined in MIL-STD-210 and to the following specific conditions expected in worldwide ground and airborne operations:

a. Pressure altitude range from 1,300 feet below sea level to the operational altitude of the aircraft

b. Ambient air temperatures from -65°F to the established maximum temperature extremes (see 3.4.5.3)

c. Salt-fog conditions as encountered in service in the vicinity of a salt-sea atmosphere

d. Relative humidity up to 100 percent, including conditions under which condensation occurs in the form of water and frost

e. Rain as encountered under service conditions

f. Sand and dust as encountered during blowing of fine sand and dust particles in desert areas

g. Fungus conditions as encountered in tropical climates

h. Vibratory excitation and acoustic noise levels as follows:

(1) Mechanical vibration as encountered during all normal modes of handling, shipping, and service in the air vehicle installation

(2) Sound-pressure levels as encountered during aircraft operations

i. Gunfire vibration (when applicable)

j. Shock levels as encountered during all normal modes of shipping, handling, and service in the air vehicle environment

k. Acceleration as encountered in service usage.

3.4.12.1 Explosive atmosphere operation. Components and equipment within the propulsion installation system shall function satisfactorily within an explosive atmosphere without causing ignition of the atmosphere.

3.4.13 Safety. The contractor shall comply with MIL-STD-882 to the extent specified by the procuring activity (see 6.2.d).

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3.5 Finishes

3.5.1 Protective treatments, coatings, and paint finishes. Protective treatments, coatings, and paint finishes shall be in accordance with applicable specifications listed in MIL-F-7179.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination. Examination of the aircraft, the aircraft mockup (if available), and appropriate contractor-prepared reports, drawings, and other data shall be used to assist in verifying contractor compliance with the requirements of this specification. The propulsion system installation shall be examined to determine that it meets the requirements of this specification with respect to design, installation, arrangement, workmanship, dimensions, finishes, and materials and any other requirements for which visual inspection will be beneficial in supplementing the contractor's engineering, logistic, or test data.

4.3 Classification of tests. Testing of the propulsion installation system shall be classified as follows:

a. Component tests

b. Endurance tests

c. Ground and flight tests.

4.3.1 Component tests. Propulsion installation system components shall be subjected to the component tests specified under 4.6.

4.3.2 Endurance tests. The propulsion installation system shall be subjected to the endurance tests specified under 4.7.

4.3.3 Ground and flight tests. The propulsion installation system shall be subjected to the ground and flight tests specified under 4.8.

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4.4 Test conditions

4.4.1 Equipment and installations. The systems, equipment, components, etc, and, if applicable, their installations which are used in providing verification of compliance with the requirements of this specification shall be subject to the approval of the procuring activity. All propulsion installation system components shall have satisfactorily completed qualification tests, preproduction tests, or at least preliminary flight-worthiness tests prior to aircraft first flight (see 3.1).

4.4.2 Laboratory conditions. Unless otherwise specified herein or by the procuring activity, all tests shall be conducted at the prevailing ambient conditions.

4.4.3 Sequence of tests. Unless otherwise specified herein or by the procuring activity, the tests specified herein may be conducted in any order.

4.4.4 Recorded data. The data recorded shall be representative of the test and the requirement to be verified. The data considered necessary to permit satisfactory evaluation of the test results shall be determined by the manufacturer and shall be subject to the approval of the procuring activity.

4.4.4.1 Accuracy of data. The accuracy of the recorded data shall be within 2.0 percent of the maximum value obtainable during the test.

4.4.5 Instrumentation. Instrumentation shall be selected to satisfy the intent of the tests specified herein. The instrumentation shall be designed, installed, and maintained in a manner suitable for demonstrating performance and operation of each subsystem.

4.4.5.1 Calibration. Test instruments shall be calibrated prior to test initiation and at intervals sufficiently frequent to ensure the required accuracy during the test. The required accuracy shall be specified on all instrumentation lists.

4.4.6 Rejection. Any part or feature of the propulsion installation system which fails to satisfy applicable requirements during any test shall be rejected. Rejected components, equipment, etc, shall be replaced or shall be reworked to correct the defects, after which all applicable tests shall be repeated. Subsystems which are rejected due to performance deficiencies shall be redesigned or, if possible, reworked to correct the deficiency, after which all applicable tests shall be repeated. Full particulars of all rejections and of all measures taken to overcome the defect or deficiency shall be recorded. All corrective actions shall be subject to procuring activity approval.

4.4.7 Test procedures. All specified tests shall be conducted in accordance with the procedures specified herein unless an alternate procedure has been

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approved (4.4.7.1). In instances where no detailed procedure is given, the contractor may conduct the test in accordance with any appropriate procedure. The contractor's test procedure shall be defined in the test procedure report (4.5.1).

4.4.7.1 Variation of test procedures. When necessary, the test procedures may deviate from that specified herein. A brief description of the alternate procedure and data which facilitates comparison with the procedure described herein shall be included in the test procedure report (4.5.1).

4.4.8 Similarity of tests. Any test that has been satisfactorily completed on a sufficiently similar propulsion installation system or any component or subsystem contained within the system need not be repeated. Documentation which substantiates test, equipment, environment, performance, and installation similarity shall be included in the test procedure report (4.5.1). The granting of a test waiver shall be contingent on procuring activity approval of the substantiating documentation.

4.4.9 Additional tests. The procuring activity reserves the right to conduct additional tests as may be considered necessary to determine conformance to the applicable requirements specified herein.

4.5 Reports

4.5.1 Test procedure report. The manufacturer shall prepare a propulsion installation system test procedure report which defines those tests required to verify that the requirements of this specification have been satisfied (see 6.2.e). The test procedure report shall define the facilities, procedures, and data which will be associated with each test. The report shall cover all applicable levels of testing such as component qualification, system simulator tests, endurance tests, air vehicle ground and flight tests, etc. The report must be approved by the procuring activity prior to initiation of any tests.

4.5.2 Test report. When required, the contractor shall prepare a report which documents the results of the tests conducted in accordance with this specification (see 6.2.f). The report shall be generally in accordance with MIL-STD-831 and shall contain the following information:

a. Plotted or tabular data of applicable parameters shall be presented in accordance with the intent of the test procedures contained herein. The frequency of data points and the form of presentation shall be chosen to demonstrate the system performance in the best manner.

b. Information, conclusions, and recommendations that would normally be incorporated in any technical report, including a description of conditions and factors peculiar to the tests, shall be presented as applicable. Any

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unusual condition, malfunction, failure, or out-of-specification performance shall be explained in detail. Reference to design requirements shall be included as applicable.

c. Supplementary information pertinent to the tests such as aircraft, engine, and propeller type and serial numbers, test rig descriptions, drawings, diagrams, photographs, etc, shall be included as required.

4.6 Component tests. In addition to the tests specified in applicable military or approved contractor-prepared specifications, all applicable components within the propulsion installation system shall be subjected to the following fire protection and environmental tests.

4.6.1 Fire protection tests

4.6.1.1 Fireproof tests. A 24-inch by 24-inch specimen of each type of firewall employed in the propulsion installation system shall be tested in accordance with the "Fireproof Tests" specified in SAE publication ARP 1055. The burner shall produce a flame temperature of 2,000°F measured 0.25 inch in front of the test specimen.

4.6.1.2 Fire-resistance tests. Representative samples of tubes, hose assemblies, fittings, supports, and other related components which carry flammable fluids and are located in potential fire zones shall be subjected to a fire-resistance test. The test shall be as specified in the "Fire Resistance Tests" of SAE publication ARP 1055 except that the fluid flow rate shall be the lowest rate expected and the fluid temperature shall be the maximum expected to be encountered during aircraft operation.

4.6.2 Environmental tests. Except as specified below, all components within the propulsion installation system shall be subjected to the following tests in order to demonstrate their compatibility with the environmental requirements applicable to their location within the propulsion installation system. Those components which undergo environmental tests specified in applicable military or approved contractor-prepared specifications need not be subjected to the following tests provided the procedures contained in the specification are equivalent in scope and severity to those contained herein. In the event of conflict between the test procedures contained herein and those contained in another specification, the following shall govern.

4.6.2.1 High altitude. High altitude tests shall be conducted in accordance with method 500, procedure II of MIL-STD-810. A pressure altitude equivalent to the operational altitude of the aircraft shall be used. While at this altitude, the component shall be operated in its normal functional mode. It shall be cycled through a pressure change from 1,300 feet below sea level to the operational altitude and returned to sea level for five cycles. Subsequent

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examination shall reveal no condensed moisture within the component or evidence of electrical breakdown.

4.6.2.2 Low temperature. Low temperature tests shall be conducted in accordance with method 502 of MIL-STD-810. The components shall meet the operational requirements of the aircraft and shall show no degradation during or upon completion of this test.

4.6.2.3 Combined low temperature and altitude. Components shall be tested at altitude and low temperature conditions simultaneously and shall meet the same requirements specified for the component in the uncombined tests.

4.6.2.4 High temperature. High temperature tests shall be conducted in accordance with method 501, procedure II of MIL-STD-810. The applicable test temperatures shall be those established in accordance with 3.4.5.3. If the cooling tests required in 4.8.5 reveal that the established temperature limits are deficient and cannot be reduced through cooling system modifications or improvement, the contractor shall conduct additional component tests at the maximum temperature extremes identified during the cooling tests of 4.8.5.

4.6.2.5 Temperature shock. Temperature shock tests shall be conducted in accordance with method 503 of MIL-STD-810. During the first few minutes after transferring the test item to the chamber with the high or low temperature, the component shall be operated in its normal operating mode.

4.6.2.6 Salt fog. Salt fog tests shall be conducted in accordance with method 509 of MIL-STD-810. The test period shall be 50 hours. Within 1 hour after compliance with this test procedure, the component shall be operated in its normal mode and shall demonstrate satisfactory operation. There shall be no corrosion or other defect that affects the operation or structural characteristics of the component.

4.6.2.7 Humidity. Humidity tests shall be conducted in accordance with method 507, procedure I of MIL-STD-810. At the conclusion of the test exposure period, the component shall be removed from the chamber and operated in its normal mode. Following the above operation, the component shall be inspected for deterioration. There shall be no corrosion or other defects that affect the operation or the structural characteristics of the component.

4.6.2.8 Rain. Rain tests shall be conducted in accordance with method 506 of MIL-STD-810. At the conclusion of the test exposure period, the component shall be removed from the chamber and operated in its normal operating mode. There shall be no evidence of corrosion or other defect that affects the operational capabilities of the component.

4.6.2.9 Dust. Dust tests shall be conducted in accordance with method 510 of MIL-STD-810. Immediately after completion of the test, the component shall

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be removed from the chamber and operated in its normal mode. Accumulated sand or dust within the component shall have no detrimental effects upon its operation and shall not be allowed in locations where excessive wear or failure might result. There shall be no damage to the exterior protective finishes.

4.6.2.10 Fungus resistance. Fungus tests shall be conducted in accordance with method 508 of MIL-STD-810. Immediately after completion of the test, the component shall be removed from the chamber and operated in its normal mode of operation. There shall be no detrimental effects upon operation of the unit. The component shall then be disassembled and inspected. There shall be no fungus on any detail of the unit where progressive damage could occur.

4.6.2.11 Vibration. Vibration tests shall be conducted in accordance with method 514 of MIL-STD-810. The equipment category shall be commensurate with the aircraft, and the test curves shall be commensurate with the mounting of the component within the system. The components shall be operated during the tests.

4.6.2.12 Gunfire vibration. Gunfire vibration tests shall be conducted on applicable components in accordance with method 519 of MIL-STD-810. Criteria for establishing component applicability is contained in the test method.

4.6.2.13 Shock. Shock tests shall be conducted in accordance with method 510, procedure I of MIL-STD-810. A sawtooth shock pulse shall be used. When appropriate, the component shall be operated during the tests.

4.6.2.14 Explosive atmosphere. Explosive atmosphere tests shall be conducted on applicable components in accordance with method 511, procedure I of MIL-STD-810.

4.6.2.15 Acceleration test. Acceleration tests shall be conducted in accordance with method 513 of MIL-STD-810. During procedure II, component functional tests shall be conducted to ensure that satisfactory component performance is obtained. There shall be no mechanical failure or malfunction due to the applied acceleration. The forward acceleration g levels shall be consistent with the aircraft performance.

4.7 Endurance tests. A 150-hour propulsion system endurance test shall be conducted to verify the integrity, compatibility, and performance of the entire propulsion system installation. The test shall be performed to detect areas of incompatibility and performance degradation due to long-term system operation and mechanical, sonic, and thermal fatigue. Examples of areas which should be investigated during this test are thrust reverser fail-safe characteristics, engine ingestion of exhaust gases due to thrust reverser operation, exhaust gas impingement on airframe surfaces, exhaust distortion effects on engine turbine blade or fan blade vibratory stresses, effects of engine bleed air duct failure on engine compartment or nacelle and pylon cowling, etc. This

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test may at the option of the contractor serve as verification of compliance with various system requirements contained in this specification provided that demonstration of those requirements is not required by some other test specified herein. The test procedure report (see 4.5.1) shall define those requirements which will be verified during the endurance test.

4.7.1 Test rig. The propulsion installation hardware and the engine(s) shall be representative of the production items and shall consist of the following systems and equipments, as applicable: the engine and thrust reverser; all engine-driven aircraft accessories; fire detection, overheat, and extinguishing systems; both the primary and backup starting systems; all nacelle and installation components (including cables) of the engine throttle and exhaust/thrust reverser control systems; the engine compartment or nacelle and pylon; and all other systems and components contained within the propulsion installation system.

4.7.2 Test procedure. The test shall consist of cycles which simulate, insofar as possible, typical aircraft missions and shall include typical accessory loading and thrust reverser operation (if applicable). Engine starting shall be demonstrated with the primary and backup starting systems and controls and shall include use of the aircraft auxiliary power unit and ground support equipment as applicable. Engine power setting distribution and test cycles used during the test shall be substantially in accordance with that specified in the paragraph entitled "Qualification Tests Procedure" of MIL-E-5009.

4.7.3 Post-test cycle inspection. Upon completion of each test cycle, the engine compartment or nacelle and pylon shall be opened and inspected during the shutdown period between cycles. The inspection shall simulate those inspections and routine maintenance actions normally performed after each flight of the aircraft. The contractor shall establish "pass/fail" criteria to be employed during the inspections. The criteria shall include requirements governing redesign and rework of systems and components due to failure or unsatisfactory operating conditions and characteristics. Any failure or condition which adversely affects aircraft safety, performance, or scheduled maintenance and availability shall require redesign and retest to verify the adequacy of the correction. The "pass/fail" criteria shall be included in the detailed test procedure report required by 4.5.1. During the course of the test, a log shall be kept of each inspection. The log shall contain a record of the general condition of the major components and documentation of all deficiencies noted during the inspection. The applicable corrective actions shall be defined. The log shall also contain a record of all maintenance performed during the shutdown period.

4.7.4 Teardown inspection. Upon completion of the 150-hour test, a teardown inspection of the engine(s) and all appropriate airframe-furnished components shall be accomplished.

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4.8 Ground and flight tests. The following ground and flight tests shall be conducted on the complete installed aircraft propulsion installation system to verify that the subsystems and specialty areas identified in the following paragraphs satisfy the requirements of this specification.

4.8.1 Engine starting. The engine starting subsystem shall be subjected to the tests specified under the paragraph entitled "Starting System" in MIL-T-25920.

4.8.2 Propulsion system controls. The propulsion system controls covered by this specification shall be subjected to the tests specified under the paragraph entitled "Power Lever Controls" of MIL-T-25920. Additional tests shall be conducted to verify control subsystem compliance with those operational and force requirements applicable to the installed subsystem which are not covered by component testing or by the tests specified in MIL-T-25920. The procedures applicable to these additional tests shall be defined in the test procedure report (see 4.5.1).

4.8.3 Fire extinguishing. The propulsion installation system fire extinguishing systems shall be subjected to tests in accordance with the paragraph entitled "Distribution Test" of MIL-E-22285.

4.8.4 Drains. Propulsion installation system drains which, due to their configuration or location, type and quantity of fluid drained, etc, appear questionable, insofar as adequate drainage is concerned, shall be subjected to drainage tests. The drainage tests shall be conducted with dyed fluids and shall verify that the performance of all questionable and potentially hazardous drains is acceptable at all applicable aircraft ground and inflight operating attitudes and conditions. The drains which will require testing and the associated test procedures shall be identified in the test procedure report (see 4.5.1).

4.8.5 Cooling and ventilation. Cooling and ventilation tests shall be conducted in accordance with the procedures specified under the paragraph entitled "Cooling" of MIL-T-25920. The allowable temperature limits shall be those established under 3.4.5.3.

4.8.6 Engine exhaust. Exhaust system tests shall be conducted in accordance with the applicable procedures specified under the paragraph entitled "Exhaust and Fan Duct and Reverser" of MIL-T-25920.

4.8.7 Air induction. Air induction system tests shall be conducted in accordance with the applicable procedures specified under the paragraph entitled "Engine Air Induction" of MIL-T-25920. Additional tests shall be conducted to verify the adequacy of the applicable air induction system protection features; e.g., inlet ice protection, FOD protection, and sand and dust protection. The procedures applicable to the additional tests shall be defined in the test procedure report (see 4.5.1).

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4.8.8 Plumbing and wiring installation. The propulsion installation system shall be subjected to chafing inspections throughout the aircraft flight and ground test program.

4.8.8.1 Chafing inspection. The plumbing and wiring installation (fuel, hydraulic, bleed air, drain, and vent lines; wiring bundles; control cabling; etc) shall be inspected to determine the following:

a. Evidence of the lack of structural adequacy of the tubing, cabling, and wire bundle mounting provisions

b. Evidence of tubing, cabling, and wiring bundle deflection, interference, and resonance

c. Evidence of connector and fitting looseness.

All instances of chafing and corrective actions shall be documented in the test report (see 4.5.2).

4.8.8.2 Inspection intervals. Unless otherwise specified by the procuring activity, the inspections shall be conducted at 50-hour intervals during the flight test program. When appropriate, additional inspections shall be conducted after the following flight conditions and maneuvers, as applicable:

a. Simulated and actual weapon delivery

b. Internal gun firing

c. Inflight refueling

d. Maximum positive and negative g maneuvers

e. Maximum roll and sideslip maneuvers

f. Maximum climbs and descents

g. Critical speed brake or thrust reverser actuations.

4.8.9 Accessibility and interchangeability. Tests for EBU and cowling interchangeability and EBU removal and replacement time and engine installation accessibility shall be conducted in accordance with the applicable procedures specified under the paragraph entitled "Engine, Propeller, Rotor, Fan, and Accessories Interchangeability" of MIL-T-25920. The tests shall determine the number of men and man-hours and elapsed time required for inspection, cleaning, and adjustment of the items listed under 3.4.10.1 and for removal and replacement of the items listed under 3.4.10.2.

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5. PREPARATION FOR DELIVERY

5.1 This section is not applicable.

6. NOTES

6.1 Intended use. This specification defines the design requirements applicable to the propulsion installations system associated with the gas turbine propulsion system of all Air Force aircraft.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification

b. Type of aircraft and particular engine installation

c. General plumbing and wiring installation specification to be submitted by the manufacturer (see 3.4.8.3)

d. Applicability of each provision of MIL-STD-882 (see 3.4.13)

e. Test procedure report to be submitted by the manufacturer for approval by the procuring activity prior to the start of any tests (see 4.5.1)

f. Test report to be submitted by the manufacturer for approval by the procuring activity (see 4.5.2).

6.3 If a manufacturer desires to employ an integrated fire and overheat detection system similar to that described in technical report AFAPL-TR-67-129, or a pyrotechnic gas generator operated fire extinguisher container similar to that described in technical report AFAPL-TR-69-66, a critical item development specification applicable to the proposed system should be prepared in accordance with MIL-STD-490.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activities:
Air Force - 71, 82

Project No. 2840-F004

User activity:
Air Force - 84

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